/ Record definition

Record definition

The fundamental unit of the format is a data record. A time series is commonly stored and exchanged as a sequence of these records. There is no interdependence of records, each is independent. There are data encodings for integers, floats, text or compressed data samples. To limit problems with timing system drift and resolution in addition to practical issues of subsetting and resource limitation for readers of the data, typical record lengths for raw data generation and archiving are recommended to be in the range of 256 and 4096 bytes.

Record layout and fields

A record is composed of a header followed by a data payload. The byte order of binary fields in the header must be least significant byte first (little endian).

The total length of a record is variable and is the sum of 40 (length of fixed section of header), field 10 (length of identifier), field 11 (length of extra headers), field 12 (length of payload).

Field	Description	Type	Length	Offset	Content
1	Record header indicator	CHAR	2	0	ASCII 'MS'
2	Format version	UINT8	1	2	Value of 3
3	Flags	UINT8	1	3	
	Record start time				
4a	Nanosecond (0 - 999999999)	UINT32	4	4	
4b	Year (0-65535)	UINT16	2	8	
4c	Day-of-year (1 - 366)	UINT16	2	10	
4d	Hour (0 - 23)	UINT8	1	12	
4e	Minute (0 - 59)	UINT8	1	13	
4f	Second (0 - 60)	UINT8	1	14	
5	Data payload encoding	UINT8	1	15	Data Encodings
6	Sample rate/period	FLOAT64	8	16	
7	Number of samples	UINT32	4	24	
8	CRC of the record	UINT32	4	28	

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Field	Description	Type	Length	Offset	Content
9	Data publication version	UINT8	1	32	
10	Length of identifier	UINT8	1	33	
11	Length of extra headers	UINT16	2	34	
12	Length of data payload	UINT32	4	36	
13	Source identifier	CHAR	V	40	URI identifier
14	Extra header fields	JSON	V	40 + field 10	
15	Data payload	encoded	V	40 + field 10 + field 11	

All length values are specified in bytes, which are assumed to be 8-bits in length. Data types for each field are defined as follows:

CHAR: ASCII encoded character data.
UINT8: Unsigned 8-bit integer.
UINT16: Unsigned 16-bit integer (little endian byte order).
UINT32: Unsigned 32-bit integer (little endian byte order).
FLOAT64: IEEE-754 64-bit floating point number (little endian byte order).
JSON: JSON Data Interchange Standard.

Description of record fields

1: CHAR: **Record header indicator**. Literal, 2-character sequence “MS”, ASCII 77 and 83, designating the start of a record.

2: UINT8: **Format version**. Set to 3 for this version. When a non-backwards compatible change is introduced the version will be incremented.

3: UINT8: **Flags**. Bit field flags, with bits 0-7 defined as:

0. Calibration signals present. [same as SEED 2.4 FSDH, field 12, bit 0]
1. Time tag is questionable. [same as SEED 2.4 FSDH, field 14, bit 7]
2. Clock locked. [same as SEED 2.4 FSDH, field 13, bit 5]
3. Reserved for future use.
4. Reserved for future use.
5. Reserved for future use.
6. Reserved for future use.
7. Reserved for future use.

latest

4: **Record start time**, time of the first data sample. A representation of UTC using individual fields for:

a. nanosecond
b. year
c. day-of-year
d. hour
e. minute
f. second

A 60 second value is used to represent a time value during a positive leap second. If no time series data are included in this record, the time should be relevant for whatever headers or flags are included.

5: UINT8: **Data payload encoding**. A code indicating the encoding format, see Section 4: Data encoding codes for a list of valid codes. If no data payload is included set this value to 0.

6: FLOAT64: **Sample rate/period**. Sample rate encoded in 64-bit IEEE-754 floating point format. When the value is positive it represents the rate in samples per second, when it is negative it represents the sample period in seconds. Creators should use the negative value sample period notation for rates less than 1 samples per second to retain resolution. Set to 0.0 if no time series data are included in the record.

7: UINT32: **Number of samples**. Total number of data samples in the data payload. Set to 0 if no samples (header-only records) or unknown number of samples (e.g. for opaque payload encoding).

8: UINT32: **CRC of the record**. CRC-32C (Castagnoli) value of the complete record with the 4-byte CRC field set to zeros. The CRC-32C (Castagnoli) algorithm with polynomial 0x1EDC6F41 (reversed 0x82F63B78) to be used is defined in RFC 3309, which further includes references to the relevant background material.

9: UINT8: **Data publication version**. Values should only be considered relative to each other for data from the same data center. Semantics may vary between data centers but generally larger values denote later and more preferred data. Recommended values: 1 for raw data, 2+ for revisions produced later, incremented for each revision. A value of 0 indicates unknown version such as when data are converted to miniSEED from another format. Changes to this value for user-versioning are not recommended, instead an extra header should be used to allow for user-versioning of different derivatives of the data.

10: UINT8: **Length of identifier**. Length, in bytes, of source identifier in field 15.

11: UINT16: **Length of extra headers**. Length, in bytes, of extra headers in field 14. If no extra headers, set this value to 0.

12: UINT32: **Length of data payload**. Length, in bytes, of data payload starting in field 15. If no data payload is present, set this value to 0. Note that no padding is permitted in the data record itself, although padding may exist within the payload depending on the type of encoding used.

13: CHAR: **Source identifier**. A unique identifier of the source of the data contained in the record. Recommended to use URI-based identfiers. Commonly an FDSN Source Identifier.

14: JSON: **Extra header fields**. Extra fields of variable length encoded in JavaScript Object Notation (JSON) Data Interchange Standard as defined by ECMA-404. It is strongly recommended to store compact JSON, containing no non-data white space, in this field to avoid wasted space.

A reserved set of headers fields is defined by the FDSN, see Extra Headers. Other header fields may be present and should be defined by the organization that created them.

15: encoded: **Data payload**. Length indicated in field 12, encoding indicated in field 5.